

Association of Genetic Predisposition to Inflammation With Cancer-Related Cognitive Impairment and Fatigue in Women Who Received Chemotherapy for Nonmetastatic Breast Cancer

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ABSTRACT

PURPOSE Cancer-related cognitive impairment (CRCI) and cancer-related fatigue (CRF) are reported by approximately 75% of patients receiving chemotherapy for breast cancer and both have been linked to inflammation. We sought to test whether an inflammation polygenic risk score (iPRS) might be associated with CRCI and/or CRF.

METHODS Using data from the UK Biobank, we developed an iPRS for the INFLA-Score, a composite measure of C-reactive protein, white cell count, platelet count, and neutrophil-lymphocyte ratio. The iPRS was evaluated for association with CRCI and CRF among women with nonmetastatic breast cancer enrolled in two completed multisite clinical trials: URCC08106 and URCC10055. CRCI and CRF were measured before and after standard-of-care chemotherapy using the FACT-Cog and Multidimensional Fatigue Symptom Inventory–Short Form (MFSI-SF), respectively. Linear regression evaluated the change in FACT-Cog and MFSI scores; logistic regression evaluated binary outcomes of any versus no worsening of scores. Analyses were adjusted for patient and treatment factors. We also performed exploratory genome-wide analyses.

RESULTS The cohort included 802 women who received chemotherapy (anthracycline-based = 51.8%; previous surgery = 85.0%) at a median age of 55 years (range = 22–81). The iPRS was associated with a significant decrease in MFSI-SF score ($\beta = -3.29$ [95% CI, -6.25 to -0.34]; $P = .029$) and lower odds of decreased MFSI-SF score (odds ratio, 0.66 [95% CI, 0.47 to 0.93]; $P = .016$) following chemotherapy. This negative relationship was partially explained by a positive correlation of the iPRS with prechemotherapy MFSI-SF score ($\beta = 4.33$ [95% CI, 0.23 to 8.43]; $P = .038$). The iPRS was not associated with a change in FACT-Cog score ($\beta = -1.12$; $P = .627$), but single nucleotide polymorphism rs9365961 was ($\beta = -10.05$; $P = 1.46 \times 10^{-8}$).

CONCLUSION If validated, the iPRS could identify patients in need of supportive care interventions to reduce CRF.

ACCOMPANYING CONTENT

 [Data Sharing Statement](#)

 [Data Supplement](#)

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INTRODUCTION

Cancer-related cognitive impairment (CRCI) and cancer-related fatigue (CRF) are reported by approximately 75% of patients receiving chemotherapy for breast cancer and may persist for years following treatment.^{1–4} These toxicities can reduce mental and physical functioning, result in mood disturbances, and interfere with daily activities including those necessary for completing cancer treatment,

performing at work, and caring for and socially interacting with family members.^{5–7} They adversely affect quality of life and recovery.^{8–10}

Systemic inflammation may contribute to the development of CRCI and CRF and may exacerbate the effects of systemic treatments such as chemotherapy, which increase CRCI and CRF.^{8,11–13} Previous studies have shown associations between CRF and CRCI with levels of circulating cytokines

CONTEXT

Key Objective

Does genetic predisposition to low-grade systemic inflammation influence the likelihood of developing cancer-related cognitive impairment (CRCI) or fatigue (CRF) following chemotherapy for nonmetastatic breast cancer?

Knowledge Generated

We developed a novel inflammation polygenic risk score (iPRS) using the INFLA-Score, a composite measure of blood biomarkers of inflammation, and tested its association with CRCI and CRF. A higher iPRS was associated with worse CRF before chemotherapy and an improvement in CRF following chemotherapy. The iPRS was not associated with CRCI, but individual genetic variants near the *TBXT* gene were.

Relevance

The iPRS could be used provide clarity to patients about their CRF-related symptoms and manage expectations over the course of cancer treatment. It may enable clinicians to direct relevant interventions, for example, pharmacological, dietary, or behavioral, that may ameliorate CRF symptoms.

including IL-6 and its receptor, TNF- α , and C-reactive protein (CRP), as well as blood cell counts; however, results across studies have been inconsistent, possibly owing to the fluctuating nature of such markers over time and depending on the disease status, comorbidities, and other external exposures.^{8,11–13}

Chronic, low-grade systemic inflammation (LGSi) is present among individuals in the general population and is a polygenic trait with common single nucleotide polymorphisms (SNPs) associated with levels of circulating biomarkers of inflammation.^{14,15} Previous studies of genetic predisposition to CRCI and CRF in patients who received chemotherapy for breast cancer focused on individual SNPs within candidate genes selected on the basis of previous links to cognitive impairment, memory impairment, or CRCI^{16–20}; immune and inflammatory pathways²¹; or the dopamine pathway, pain regulation, circadian regulation, and blood-brain barrier drug efflux transporters.²²

We aimed to capture genetic predisposition to LGSi more broadly and examine associations with CRCI and CRF. We created an inflammation polygenic risk score (iPRS) using the INFLA-Score, a composite measure of LGSi based on circulating levels of biomarkers of inflammation including CRP, WBC count, platelet count, and neutrophil-lymphocyte ratio (NLR),²³ and assessed the association between the iPRS and the development of CRCI and CRF in women who received chemotherapy for nonmetastatic breast cancer (NMBC). Additionally, we performed an exploratory genome-wide association study (GWAS) of CRCI and CRF to identify risk SNPs outside of known inflammation genes/pathways. Understanding the influence of genetic predisposition to LGSi on the likelihood of developing CRCI and CRF in patients who receive chemotherapy may shape our approach to care.

METHODS

Study Population

We performed a secondary analysis of two clinical studies from the University of Rochester Cancer Center (URCC) NCI Community Oncology Research Program (NCORP) Research Base and affiliates: URCC10055²⁴ and URCC08106²⁵ (details in Supplemental Methods). This study was approved by the institutional review boards of the University of Rochester and the Medical College of Wisconsin, and all participants provided written informed consent to participate in URCC10055/URCC08106 and for their data to be included in secondary analyses of cancer-related outcomes.

CRCI and CRF Outcomes

CRCI was measured using the Functional Assessment of Cancer Therapy–Cognitive Function (FACT-Cog) version 2,²⁶ and CRF was measured using the Multidimensional Fatigue Symptom Inventory–Short Form (MFSI-SF).²⁷ We analyzed the change in FACT-Cog and MFSI scores from prechemotherapy to approximately 6 months post-chemotherapy. A decrease in the FACT-Cog score represents a decline in cognitive function. Conversely, a higher MFSI-SF score represents worsening fatigue.

Germline SNP Data in URCC10055/URCC08106

Genome-wide SNP data were generated with Illumina Global Diversity Arrays using germline DNA isolated from whole blood collected at the time of enrollment on URCC10055/URCC08106. Standard quality control filters were applied (removal of samples with >5% missing genotypes, SNPs missing in >5% of samples, and samples with >40% or <5% heterozygosity), and SNP imputation was performed using

the TOPMed server.^{28,29} Genetic analyses include 783 participants with high-quality data available.

Development of an INFLA-Score Polygenic Risk Score (iPRS) in the UK Biobank

The INFLA-Score²³ is a composite measure including WBC count, platelet count, NLR, and serum CRP, with standardized values ranging from -16 to 16. We conducted a GWAS of the INFLA-Score among 372,299 unrelated participants with European ancestry in the UK Biobank with nonmissing data for the four biomarkers (details in Supplemental Methods).³⁰ Each SNP was tested for association with INFLA-Score using linear regression adjusted for age, sex, and 10 principal components (provided by the UK Biobank) to control for genetic ancestry in Plink-2.0.^{31,32} PRSice-2 was used to perform linkage-disequilibrium-based clumping in the UK Biobank (250 kb blocks, r^2 threshold 0.1), and the top SNP within each clump meeting genome-wide significance ($P < 5 \times 10^{-8}$) was selected for inclusion in the iPRS. The iPRS was internally validated in the UK Biobank (Supplemental Methods).

The iPRS was calculated for each participant in URCC10055/URCC08106 by summing the number of effect alleles for each SNP weighted by the regression coefficient from the UK Biobank GWAS.

Statistical Analysis

Continuous variables were reported as median/range and compared between cohorts using Wilcoxon rank-sum tests; categorical variables were reported as frequency/percentage and compared using chi-square tests. Univariable linear regression estimated the relationship between baseline characteristics and the change in FACT-Cog and MFSI-SF scores separately. Variables with $P < .10$ were included in subsequent multivariable models, which included participants with nonmissing data for all covariates, fit using Stata/SE 18.0.

Multivariable linear regression estimated the relationship between the iPRS and each of the change in FACT-Cog score and the change in MFSI-SF score from prechemotherapy to postchemotherapy adjusted for identified covariates. Multivariable logistic regression secondarily evaluated a binary outcome of CRF defined as any increase in MFSI-SF versus no increase.

Exploratory GWAS was performed for each of CRCI and CRF in the URCC10055/URCC08106 cohort. The GWAS included SNPs with MAF >5% and imputation info >0.3 and was adjusted for baseline characteristics identified as described above and the first three principal components.

RESULTS

URCC10055/URCC08106 Cohort Characteristics

The URCC10055/URCC08106 cohort includes 802 women who received chemotherapy for NMBC (Table 1). The median age was 55 years (range, 22–81 years), and most participants were postmenopausal (53.9%) and underwent previous surgery (85%) with approximately half (51.8%) having received anthracycline-based chemotherapy. Both cohorts were largely similar in characteristics except for statistically significant differences in age, ethnicity, marital status, previous surgery, KPS, and antihuman epidermal growth factor receptor 2 therapy (Table 1). As expected, fewer participants in URCC08106 engaged in exercise before chemotherapy based on study inclusion criteria.

There was a median decrease (worsening) pre- to postchemotherapy in the FACT-Cog score of -9 (range, -141 to 74), with 64.1% of the women reporting any decrease. Similarly, there was a median increase (worsening) in the MFSI-SF score of 7.0 (range, -64 to -71), with 63.2% of the women reporting any increase. Factors associated with the change in FACT-Cog included age, education, menopausal status, and prechemotherapy FACT-Cog (Data Supplement, Table S1). Factors associated with the change in MFSI-SF included BMI, education, previous surgery, and prechemotherapy MFSI-SF (Data Supplement, Table S2).

Development of the iPRS and Association With CRCI and CRF

The UK Biobank GWAS of the INFLA-Score revealed high polygenicity (Data Supplement, Fig S1), with several significant loci tagging genes with a well-characterized role in inflammation/inflammatory diseases including CRP (chr1q23.2), APOE (chr19q13.32), and the human leukocyte antigen locus (chr6p22.1; Data Supplement, Fig S1). After clumping and thresholding, 1,113 SNPs were selected for inclusion in the iPRS (Data Supplement, Table S3) of which 953 were available for calculation of the iPRS in the URCC10055/URCC070 cohort. As anticipated, some SNPs were unavailable due to poor imputation and/or low MAF in this cohort, which was more diverse than the UK Biobank.

There was no statistically significant association between the iPRS and the change in FACT-Cog score from pre- to postchemotherapy on univariable analysis ($\beta = -1.12$ [95% CI, -5.62 to 3.39]; $P = .627$; Data Supplement, Table S1) or after adjustment for covariates ($\beta = -.36$ [95% CI, -4.76 to 4.03]; $P = .871$; Table 2). Similarly, there was no significant association between the iPRS and a binary measure of CRCI (odds ratio [OR], 1.14 [95% CI, 0.82 to 1.58]; $P = .45$; adjusted OR, 1.13 [95% CI, 0.80 to 1.58]; $P = .491$; Data Supplement, Table S4).

TABLE 1. Characteristics of 802 Women Who Received Chemotherapy for Nonmetastatic Breast Cancer in URCC10055 and URCC08106

Patient Characteristic	All (N = 802)	URCC10055 (N = 518)	URCC08106 (N = 284)	P ^a
Age ^a (years), median (range)	55 (22-81)	54 (22-81)	56 (28-79)	.048
Race, ^a No. (%)				.041
White	694 (87.2)	455 (88.8)	239 (84.2)	
Black/African American	77 (9.7)	42 (8.2)	35 (12.3)	
Asian	20 (2.5)	10 (2.0)	10 (3.5)	
American Indian	5 (0.6)	5 (1.0)	0 (0.0)	
Ethnicity, ^b No. (%)				.68
Hispanic/Latino	12 (1.5)	7 (1.4)	5 (1.8)	
Not Hispanic/Latino	777 (98.5)	498 (98.6)	279 (98.2)	
Marital status, ^a No. (%)				.017
Single	80 (10.1)	40 (7.8)	40 (14.1)	
Married or in long-term relationship	557 (70.0)	369 (72.1)	188 (66.2)	
Divorced or separated	111 (13.9)	76 (14.8)	35 (12.3)	
Widowed	48 (6.0)	27 (5.3)	21 (7.4)	
Educational status, ^c No. (%)				.133
College or higher	594 (74.7)	393 (76.6)	201 (71.3)	
Completed high school or equivalent	184 (23.1)	112 (21.8)	72 (25.5)	
Less than high school	17 (2.1)	8 (1.6)	9 (3.2)	
BMI ^d (kg/m ²), median (range)	29 (15-65)	30 (15-65)	29 (17-57)	.856
Menopause status, ^e No. (%)				.644
Premenopausal	242 (30.5)	160 (31.3)	82 (29.1)	
Medically induced	68 (8.6)	45 (8.8)	23 (8.2)	
Perimenopausal	56 (7.1)	39 (7.6)	17 (6.0)	
Postmenopausal	428 (53.9)	268 (52.3)	160 (56.7)***	
Previous breast cancer surgery, No. (%)				.005
No	120 (15.0)	91 (17.6)	29 (10.2)	
Yes	682 (85.0)	427 (82.4)	255 (89.8)	
Tumor stage, ^f No. (%)				.642
I	217 (28.3)	135 (27.8)	82 (29.2)	
II	390 (50.9)	253 (52.2)	137 (48.8)	
III	159 (20.8)	97 (20.0)	62 (22.1)	
Karnofsky performance score, ^e No. (%)				.001
100	539 (67.9)	369 (72.4)	170 (59.9)	
90	207 (26.1)	116 (22.7)	91 (32.0)	
80	41 (5.2)	19 (3.7)	22 (7.7)	
≤70	7 (0.9)	6 (1.2)	1 (0.4)	
Anthracycline therapy, ^f No. (%)				.956
No	369 (48.2)	234 (48.2)	135 (48.0)	
Yes	397 (51.8)	251 (51.8)	146 (52.0)	
Anti-HER2 therapy, ^g No. (%)				<.001
No	184 (23.9)	143 (29.5)	41 (14.4)	
Yes	585 (76.1)	342 (70.5)	243 (85.6)	
Prechemotherapy exercise, ^a No. (%)				<.001
No	568 (71.4)	285 (55.7)	283 (99.6)	
Yes, <6 months	55 (6.9)	54 (10.5)	1 (0.4)	
Yes, more than 6 months	173 (21.7)	173 (33.8)	0 (0.0)	
Prechemotherapy FACT-Cog, ^h median (range)	167 (45-200)	165 (62-200)	174 (45-200)	<.001
Postchemotherapy FACT-Cog, ⁱ median (range)	153 (18-200)	150 (18-200)	164 (53-200)	<.001
Change in FACT-Cog, ^j median (range)	-9 (-141 to 74)	-10.5 (-141 to 74)	-5.5 (-108 to 51)	.028

(continued on following page)

TABLE 1. Characteristics of 802 Women Who Received Chemotherapy for Nonmetastatic Breast Cancer in URCC10055 and URCC08106 (continued)

Patient Characteristic	All (N = 802)	URCC10055 (N = 518)	URCC08106 (N = 284)	P ^a
Any decrease in FACT-Cog, ^j No. (%)				.120
No	251 (35.9)	160 (33.9)	91 (39.9)	
Yes	449 (64.1)	312 (66.1)	137 (60.1)	
Prechemotherapy MFSI-SF, ^k median (range)	4 (–24 to 83)	5 (–24 to 83)	2 (–23 to 67)	.051
Postchemotherapy MFSI-SF, ^l median (range)	12 (–24 to 88)	14 (–24 to 88)	9 (–24 to 72)	.004
Change in MFSI-SF, ^m median (range)	7.0 (–64 to 71)	8.0 (–64 to 71)	5.0 (–31 to 61)	.053
Any increase in MFSI-SF, ^m No. (%)				.085
No	257 (36.8)	162 (34.6)	95 (41.3)	
Yes	441 (63.2)	306 (65.4)	135 (58.7)	

Abbreviations: HER2, human epidermal growth factor receptor 2; MFSI-SF, Multidimensional Fatigue Symptom Inventory–Short Form; URCC, University of Rochester Cancer Center.

^aAge, race, marital status, and prechemotherapy exercise were missing for six participants.

^bEthnicity was missing for 13 participants.

^cEducational status was missing for seven participants.

^dBMI was missing for 14 participants.

^eMenopause status and KPS were missing for eight participants.

^fTumor stage, anthracycline therapy were missing for 36 participants.

^gAnti-HER2 therapy was missing for 33 participants.

^hPrechemotherapy FACT-Cog score was missing for 16 participants.

ⁱPostchemotherapy FACT-Cog score was missing for 96 participants.

^jChange in FACT-Cog score was missing for 102 participants.

^kPrechemotherapy MFSI-SF score was missing for 17 participants.

^lPostchemotherapy MFSI-SF score was missing for 99 participants.

^mChange in MFSI-SF score was missing for 104 participants.

ⁿThe characteristics of both cohorts were compared using the Wilcoxon rank-sum test for continuous variables and the chi-square test for categorical variables. Participants with missing data for a given variable were excluded from analysis involving the variable. Bold values indicate $P < .05$.

There was a statistically significant negative association between the iPRS and change in MFSI from pre- to postchemotherapy ($\beta = -3.29$ [95% CI, -6.25 to -0.34]; $P = .029$; Data Supplement, Table S2) which was only slightly attenuated in a multivariable model ($\beta = -2.71$ [95% CI, -5.50 to 0.08]; $P = .057$; Table 3). The relationship was stronger with a binary measure of MFSI increase (OR, 0.66 [95% CI, 0.47 to 0.93]; $P = .016$; covariate-adjusted OR, 0.67 [95% CI, 0.47 to 0.96]; $P = .029$, Data Supplement, Table S5). These results indicate that women with a *higher* iPRS, indicative of greater genetic predisposition to LGSI, are *less likely* to report worsening of fatigue over the course of chemotherapy compared with women who have a *lower* iPRS. This inverse relationship was confirmed when the iPRS was grouped into quartiles where women with an iPRS in the *lowest* quartiles had significantly increased odds of worsening fatigue pre- to postchemotherapy (Table 4, and Data Supplement, Fig S3 showing the change in MFSI).

Influence of Prechemotherapy CRF and Exercise

The inverse relationship of the iPRS with the change in fatigue might be partially explained by prechemotherapy fatigue. We found a positive correlation between the iPRS and

prechemotherapy MFSI ($\beta = 2.63$ [95% CI, -0.42 to 5.68]; $P = .091$) and a statistically significant association when comparing women with an iPRS in the highest versus lowest quartile ($\beta = 4.33$ [95% CI, 0.23 to 8.43]; $P = .038$; Table 5). With the results above, these findings suggest that a higher iPRS contributes to more fatigue before chemotherapy that does not worsen and may improve during treatment, whereas women with a lower iPRS have less fatigue before chemotherapy and are at greatest risk of developing new or worsening fatigue during treatment.

Given the known role of physical activity in modulating systemic inflammation and the relationship between exercise and CRF, we assessed whether the effect of the iPRS on CRF was modified by baseline exercise status. A stratified analysis by baseline exercise status showed a trend of higher odds of increased MFSI (ie, worsening fatigue) among those with an iPRS in lower two quartiles versus the highest quartile (Q1, OR, 2.43 [95% CI, 0.94 to 6.31]; $P = .068$; Q2, OR, 3.45 [95% CI, 1.26 to 9.42]; $P = .016$; Table 6). However, a linear model that included the iPRS, baseline exercise status, and their interaction term showed no evidence of a statistically significant interaction ($\beta = .19$ [95% CI, -7.14 to 7.52]; $P = .959$).

TABLE 2. Linear Regression of Change in FACT-Cog Score on iPRS, Adjusted for Patient- and Treatment-Related Factors in 675 Women Who Received Chemotherapy for Nonmetastatic Breast Cancer in URCC10055 and URCC08106

Independent Variable	Coefficient (95% CI)	P
iPRS	−0.36 (−4.76 to 4.03)	.871
Prechemotherapy FACT-Cog	−0.19 (−0.27 to −0.12)	<.001
Age (years)	0.54 (0.23 to 0.84)	.001
Educational status		
College or higher	Ref.	—
Completed high school or equivalent	4.06 (−0.94 to 9.06)	.111
Less than high school	8.13 (−11.39 to 27.66)	.414
Menopause status		
Premenopausal	Ref.	—
Medically induced	−8.13 (−17.12 to 0.87)	.076
Perimenopausal	−5.40 (−14.55 to 3.76)	.247
Postmenopausal	−2.30 (−9.51 to 4.91)	.531
Karnofsky performance status		
100	Ref.	—
90	−1.27 (−6.18 to 3.63)	.610
80	−9.99 (−19.83 to −0.14)	.047
≤70	−2.42 (−27.21 to 22.37)	.848
Anthracycline regimen		
No	Ref.	—
Yes	−2.03 (−6.29 to 2.22)	.349

Abbreviations: iPRS, inflammation polygenic risk score; Ref., reference; URCC, University of Rochester Cancer Center.

GWAS of CRCI and CRF

The GWAS identified a region on chr6q27, tagged by rs9365961, significantly associated with a change in FACT-

TABLE 3. Linear Regression of Change in MFSI-SF Score on iPRS, Adjusted for Patient- and Treatment-Related Factors in 675 Women Who Received Chemotherapy for Nonmetastatic Breast Cancer in URCC10055 and URCC08106

Independent Variable	Coefficient (95% CI)	P
iPRS	−2.71 (−5.50 to 0.08)	.057
Prechemotherapy MFSI	−0.31 (−0.37 to −0.24)	<.001
BMI	0.22 (0.03 to 0.41)	.021
Educational status		
College or higher	Ref.	—
Completed high school or equivalent	−2.54 (−5.69 to 0.61)	.114
Less than high school	−7.29 (−18.9 to 4.30)	.217
Previous breast cancer surgery		
No	Ref.	—
Yes	−1.59 (−5.31 to 2.13)	.401

Abbreviations: iPRS, inflammation polygenic risk score; MFSI-SF, Multidimensional Fatigue Symptom Inventory–Short Form; Ref., reference; URCC, University of Rochester Cancer Center.

TABLE 4. Logistic Regression of Increased MFSI-SF Score (v decrease or no change) on iPRS Quartile, Adjusted for Patient- and Treatment-Related Factors in Women Who Received Chemotherapy for Nonmetastatic Breast Cancer in URCC10055 and URCC08106

Independent Variable	Odds Ratio (95% CI)	P
iPRS		
Quartile 4	Ref.	Ref.
Quartile 3	1.16 (0.73 to 1.84)	.533
Quartile 2	1.70 (1.07 to 2.72)	.025
Quartile 1	1.69 (1.05 to 2.71)	.031
Prechemotherapy MFSI	0.97 (0.96 to 0.98)	<.001
BMI	1.03 (1.01 to 1.06)	.012
Prechemotherapy exercise		
No exercise	Ref.	—
Regularly exercising for <6 months	2.30 (1.07 to 4.91)	.032
Regularly exercising for >6 months	0.97 (0.65 to 1.45)	.882
Previous breast cancer surgery		
No	Ref.	—
Yes	0.73 (0.45 to 1.19)	.208

NOTE. Analysis includes 676 women.

Abbreviations: iPRS, inflammation polygenic risk score; MFSI-SF, Multidimensional Fatigue Symptom Inventory–Short Form; Ref., reference; URCC, University of Rochester Cancer Center.

Cog ($\beta = -10.05$ [95% CI, -13.48 to -6.62]; $P = 1.46 \times 10^{-8}$; Data Supplement, Fig S2A). This region is approximately 8 kb upstream of *TBXT*. rs9365961 does not overlap known regulatory sequences in ENCODE and is not an eQTL for *TBXT* in GTEx,³³ although tissues in GTEx are not chemotherapy-exposed. In addition, 83 regions showed suggestive associations with change in FACT-Cog that warrant evaluation in a larger cohort. No SNPs reached genome-wide significance for change in MFSI-SF score (Data Supplement, Fig S2B), although 104 regions showed suggestive associations, two of which approached the threshold for genome-wide significance (*rs2152390*, *chr1:17914405:T>C*; $\beta = 5.11$ [95% CI, 3.18 to 7.03]; $P = 2.68 \times 10^{-7}$; and *rs66879462*, *chrX:125179450:CA>C*; $\beta = -4.90$ [95% CI, -6.79 to -3.00]; $P = 5.23 \times 10^{-7}$) and warrant further evaluation.

TABLE 5. Linear Regression of Prechemotherapy Multidimensional Fatigue Symptom Inventory–Short Form Score on iPRS Quartile in Women Who Received Chemotherapy for Nonmetastatic Breast Cancer in URCC10055 and URCC08106 (N = 776)

Independent Variable	Coefficient (95% CI)	P
iPRS		
Quartile 1	Ref.	—
Quartile 2	3.05 (−1.06 to 7.15)	.145
Quartile 3	5.28 (1.18 to 9.38)	.012
Quartile 4	4.33 (0.23 to 8.43)	.038

Abbreviations: iPRS, inflammation polygenic risk score; Ref., reference; URCC, University of Rochester Cancer Center.

TABLE 6. Logistic Regression of Increased Multidimensional Fatigue Symptom Inventory–Short Form Score (decrease or no change) on iPRS Quartile Stratified by Baseline Exercise Status

Independent Variable	Odds Ratio (95% CI)	P
Not regularly exercising or exercising for <6 months, N = 522		
iPRS		
Quartile 4	Ref.	—
Quartile 3	1.23 (0.74 to 2.03)	.410
Quartile 2	1.27 (0.77 to 2.07)	.349
Quartile 1	1.58 (0.95 to 2.63)	.078
Regularly exercising >6 months, N = 157		
iPRS		
Quartile 4	Ref.	—
Quartile 3	0.59 (0.23 to 1.51)	.268
Quartile 2	3.45 (1.26 to 9.42)	.016
Quartile 1	2.43 (0.94 to 6.31)	.068

Abbreviations: iPRS, inflammation polygenic risk score; Ref., reference.

DISCUSSION

This study suggests there is an association between an iPRS capturing genetic predisposition to LGSI and CRF, and that exercise may influence this association. It shows that patients with a higher iPRS are likelier to report worse fatigue prechemotherapy and report an improvement in their fatigue postchemotherapy. We found no evidence that the iPRS is associated with a change in CRCI symptoms postchemotherapy.

This is the first report, to our knowledge, of the association between an inflammation-based PRS and CRF, and the first GWAS of CRF and CRCI. Previous studies of links between inflammation and CRCI or CRF focused on biomarkers such as cytokines and chemokines,^{11–13} or used a more limited approach that assessed the association with specific candidate genes.^{34–38} In contrast, the risk locus identified for CRCI via agnostic GWAS was not previously considered and may shed light on possible biologic mechanisms. *TBXT* codes for an embryonic nuclear transcription factor that affects transcription of genes required for mesoderm formation and differentiation, and the protein is localized to notochord-derived cells. Variation in this gene is associated with susceptibility to neural tube defects and chordoma.³⁹ If validated, functional characterization of this gene may be warranted.

Previous genetic studies of CRF and CRCI were limited to one or a few candidate genes. Studies of CRF evaluated candidate SNPs in genes from both inflammatory and noninflammatory pathways.^{34–37} Most of the studies were cross-sectional and most assessed fatigue before or after treatment for cancer. Some of the studies focused on breast cancer alone, whereas others included other cancer types. Of these studies, one showed an association between a genetic risk score

comprising three nonsynonymous SNPs (*ILB* -511 C>T, *rs16944*; *IL6* -174 G>C, *rs1800795*; and *TNF* -308 G>A, *rs1800629*) and fatigue within 3 months of treatment in 171 women with early-stage breast cancer.³⁸ Another found SNPs in two candidate genes outside of inflammatory pathways, *rs6277* in *DRD2* and *rs1801260* in *CLOCK*, to be associated with higher odds of fatigue among 67 patients undergoing chemotherapy for breast cancer.²² Other studies showed association between CRF and *rs6265* in brain-derived neurotrophic factor (*BDNF*) in female cancer survivors⁴⁰ and *rs25531* in *SLC6A4* in patients diagnosed with colon and rectal cancer.⁴¹ Kuehl et al analyzed 45 candidate SNPs in the inflammatory pathway genes that were previously linked to CRF and found no association with CRF at multiple time points in a cross-sectional study of patients with breast cancer. However, they found a significant association between *rs3093662* in *TNF-α* and persistent physical CRF—a subdomain of CRF.⁴² Regarding CRCI, previous genetic studies largely focused on candidate genes linked to cognitive impairment or memory impairment. For example, several studies have reported that the association of *APOE ε2* and/or *ε4* polymorphisms, associated with a risk of Alzheimer's disease, had similar effect on CRCI.^{43,44} Yap et al⁴⁵ showed that *rs6265* in *BDNF*, also linked to Alzheimer's disease, was protective against CRCI in certain domains of the FACT-Cog questionnaire among survivors of early-stage breast cancer. In contrast, Hsiao et al¹⁹ found *rs6265* to be associated with lower self-reported cognitive abilities in a cross-sectional study of 353 breast cancer survivors. Hajj et al²⁰ found no significant association between *rs197791* in the opioid receptor gene *OPRM1* and FACT-Cog score in a cross-sectional study of 112 women who received chemotherapy for breast cancer, although they found an association with higher quality-of-life subdomain of the FACT-Cog. Outside of pathways related to neurocognition, Iranzo et al¹⁶ found *rs1800795* near *IL-6* to be associated with cognitive decline following completion of chemotherapy in patients with breast and colon cancer.

The finding of an association between the iPRS and CRF supports previous studies showing a link between inflammation and the development of CRF that used circulating biomarkers and extends these findings to show there is an inherited genetic basis.⁸ Fatigue levels have been positively associated with high-sensitivity CRP and WBC count in disease-free breast cancer survivors.⁴⁶ However, it is noteworthy that the higher predisposition to LGSI is associated with improvement in fatigue symptoms following chemotherapy. A possible explanation is that the higher level of prechemotherapy fatigue is due in part to a higher inflammatory response to the tumor as cancer is associated with a proinflammatory environment.^{47,48} Thus, with chemotherapy and resulting reduced tumor burden, there may be an improvement in the inflammatory state and a subsequent improvement of fatigue. This idea is consistent with the negative association with the postchemotherapy MFISI-SF score and the positive association between the iPRS and the prechemotherapy MFISI-SF score, but such a mechanistic

hypothesis requires testing in a preclinical model and/or prospective study with serial biomarkers collected.

Understanding the relationship between genetic predisposition to inflammation and the likelihood of developing CRF may influence approach to care. Individuals with a high iPRS may benefit from therapy that targets the inflammatory process. These individuals may benefit from anti-inflammatory medications and/or prechemotherapy exercise regimen. The URCC08106 study showed exercise to be associated with low-level proinflammatory cytokines and increased anti-inflammatory cytokines in patients who received 2-week cycles of chemotherapy for NMBC.²⁵ Understanding the relationship between genetic predisposition to inflammation and the likelihood of developing CRF may also help inform patient expectations before and during treatment.

This study has notable strengths including the use of validated patient-reported outcomes to capture symptoms of CRF and CRCI pre-/postchemotherapy, systematic collection of patient-related and cancer treatment-related covariates, and inclusion of a diverse, nation-wide patient population. Limitations include heterogeneity across the two studies, lack of data on some potentially important exposures such as diet and smoking, and limited sample size for the GWAS. Similarly, because URCC08106 required participants not be in the active or maintenance stage of exercise behavior for eligibility, the sample size for the stratified analysis by exercise status was skewed toward nonexercisers. In addition, although the NCORP cohorts were diverse in terms of reported race and ethnicity, we limited the iPRS development

to European ancestry participants in the UK Biobank due to limited sample size of participants of non-European ancestry. This may limit the transferability of the iPRS, reducing statistical power to detect associations with CRCI and/or CRF. Work is underway to develop a multiancestry iPRS using emerging methods from the National Institutes of Health-supported PRIMED Consortium.⁴⁹ Finally, because limitations related to imputation accuracy can stem from heterogeneity between the cohort of interest and the reference population used for imputation, we used the latest TOPMed imputation cohort which has increased the accuracy of genotype imputation, particularly in persons of African ancestry, compared with earlier imputation panels.²⁸ Although the TOPMed panel has a limited boost in accuracy for persons of South Asian ancestry, the effect of this is limited considering self-reported Asians make up just 2.5% of our cohort.²⁸ Despite limitations, we detected a statistically significant association between the iPRS and CRF, providing insight into the influence of genetic predisposition to LGSI on CRF. It also suggests that although inflammation may influence the development of CRCI, shared genetic susceptibility to non-cancer-related cognitive phenotypes should be explored.

In conclusion, our findings suggest genetic predisposition to inflammation may influence the likelihood of developing CRF and the change in CRF during chemotherapy. This effect may be influenced by exercise prechemotherapy, pointing to a potentially modifiable risk factor. A continued understanding of the influence of these factors may shape our approach toward care of patients who receive chemotherapy for NMBC.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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Open Payments is a public database containing information reported by companies about payments made to US-licensed physicians ([Open Payments](http://OpenPayments.org)).

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